BETWEEN, IN, NOT IN, AND, OR

Lecture Notes · Game DB Junior

In this lesson

- **BETWEEN**: is a value between two numbers?
- **IN** and **NOT IN**: is a value in a list or not?
- **AND** and **OR**: use many checks in one WHERE statement

1. Three more operators

Besides simple checks like equal (=) and not equal (!=), there are three operators that do a bit more complicated checks: **BETWEEN**, **IN** and **NOT IN**.

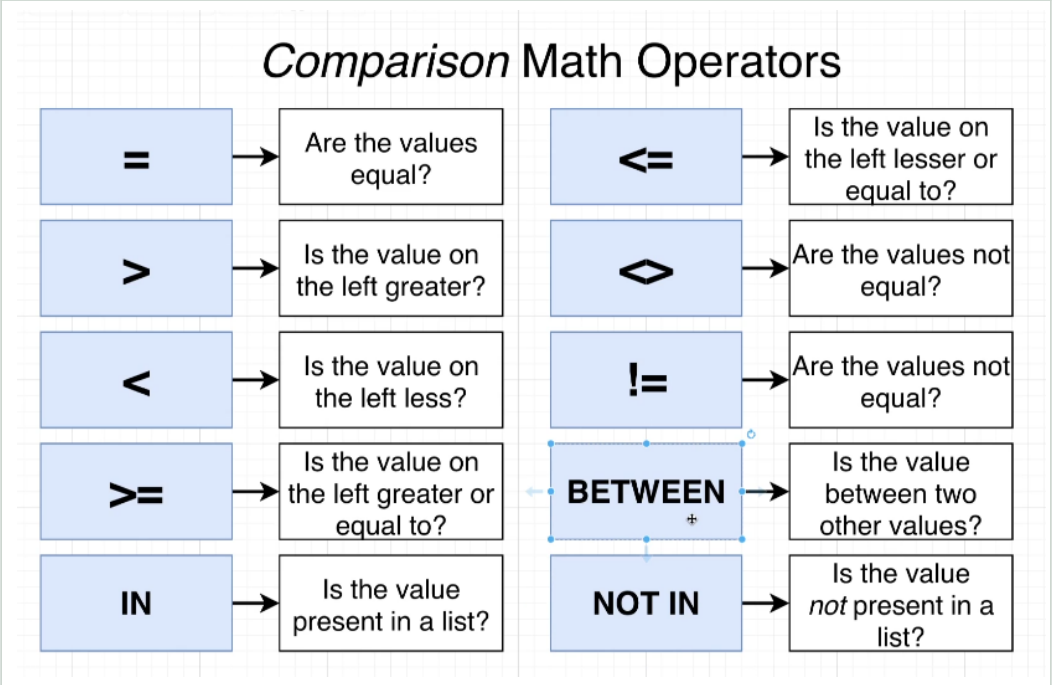


Figure 1 — Comparison math operators

2. BETWEEN

We use **BETWEEN** to check if a value is between one number and another number.

Delhi, Shanghai and Sao Paulo have areas of about 2000, 4000 and 3000. Let's find all the cities with an area between 2000 and 4000. We expect to see **Delhi** and **Sao Paulo**.

```
SELECT name, area FROM cities WHERE area BETWEEN 2000 AND 4000;
```

- How to write it: the **value** we want to look at, then **BETWEEN**, the **lower bound**, **AND**, the **upper bound**.
- When we run it, we get Delhi and Sao Paulo.
- If we make the upper bound bigger, like **5000**, Shanghai is also in the result.

3. IN

IN checks if a value is present in a **list** that we give. For now, the list is a hard coded list of values. We write the list inside **parentheses ( )**. Strings go inside single quotes.

```
SELECT name, area FROM cities WHERE name IN ('Delhi', 'Shanghai');
```

This means: give us all the cities where the name is in this list, Delhi or Shanghai. When we run it, we get just **Delhi** and **Shanghai**.

4. NOT IN

NOT IN does the exact opposite: give us all the cities that are **not** in the list.

```
SELECT name, area FROM cities WHERE name NOT IN ('Delhi', 'Shanghai');
```

We get every city that does not have the name Delhi or Shanghai: **Tokyo** and **Sao Paulo**.

Numbers in the list

We can also use **numbers** in the list, not only strings. Here we change the column from name to area:

```
SELECT name, area FROM cities WHERE area NOT IN (3043, 8223);
```

We expect all the cities except Tokyo and Sao Paulo. When we run it, we get **Delhi** and **Shanghai**.

5. Many checks: AND and OR

We can use **many operators** in one WHERE statement. This is a **compound check**.

AND

```
SELECT name, area
FROM cities
WHERE area NOT IN (3043, 8223) AND name = 'Delhi';
```

- With **AND**, a city must match **both** checks: its area is not in the list **and** its name is Delhi.
- When we run it, we see just **Delhi**.

OR

```
SELECT name, area
FROM cities
WHERE area NOT IN (3043, 8223) OR name = 'Delhi';
```

- With **OR**, a city must match **at least one** of the checks.
- We see **Delhi**, because its name is Delhi and its area is not in the list.

- We see **Shanghai**, because its area is not in the list.

Chaining many checks

We can chain **as many** AND and OR checks as we want. For example, we add one more check:

```
SELECT name, area
FROM cities
WHERE area NOT IN (3043, 8223)
    OR name = 'Delhi'
    OR name = 'Tokyo';
```

Now **Tokyo** is also in the result. WHERE statements can get quite complicated when we want to find very specific data: many small checks joined together with ANDs or ORs.

6. Running the code (my practice)

Query 1: area BETWEEN 2000 AND 4000.

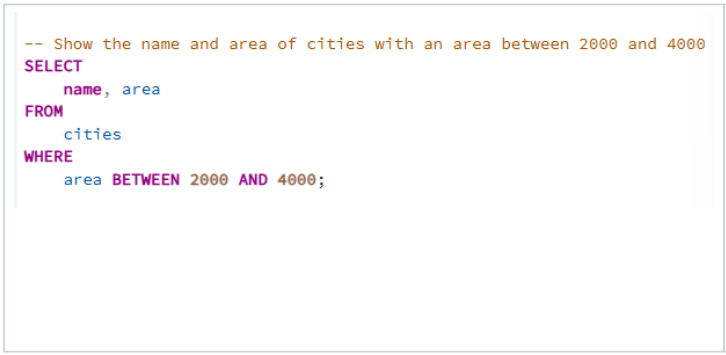


Figure 2 — BETWEEN

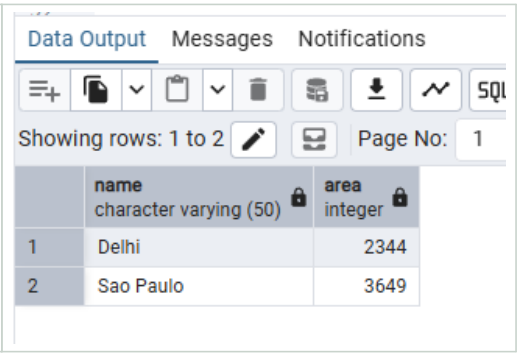


Figure 3 — Result: Delhi and Sao Paulo

Query 2: name IN ('Delhi', 'Sao Paulo').

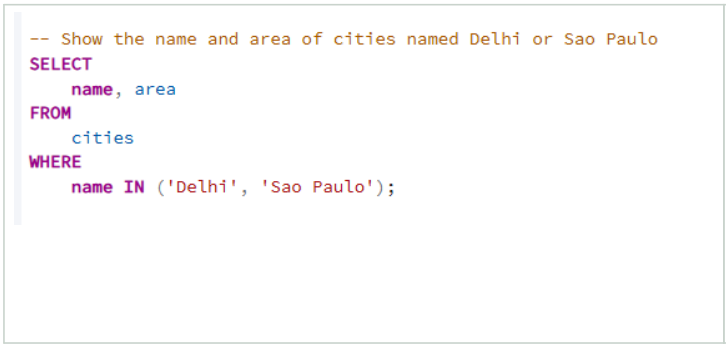


Figure 4 — IN

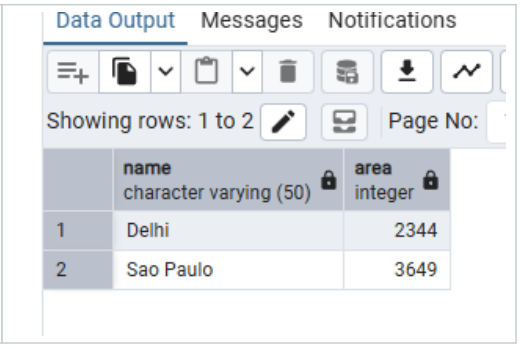


Figure 5 — Result: Delhi and Sao Paulo

Query 3: name NOT IN ('Delhi', 'Sao Paulo').

```sql
-- Show the name and area of cities except Delhi and Sao Paulo
SELECT
    name, area
FROM
    cities
WHERE
    name NOT IN ('Delhi', 'Sao Paulo');
```

Figure 6 — NOT IN

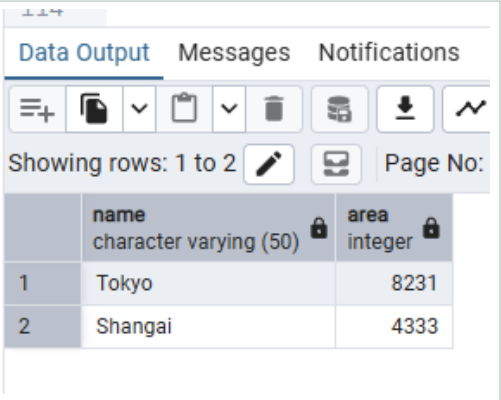


Figure 7 — Result: Tokyo and Shangai

Query 4: NOT IN with AND. A city must match both checks.

```sql
-- Show Tokyo only if it is not Delhi or Sao Paulo
SELECT
    name, area
FROM
    cities
WHERE
    name NOT IN ('Delhi', 'Sao Paulo') AND
    name = 'Tokyo';
```

Figure 8 — NOT IN ... AND ...

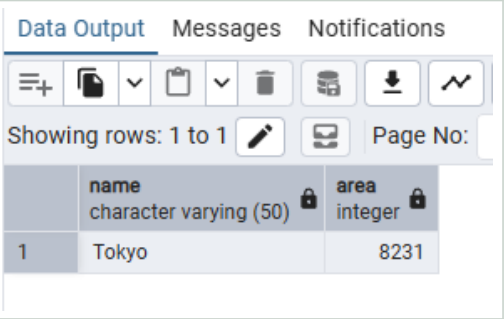


Figure 9 — Result: only Tokyo

Query 5: NOT IN with OR. A city must match at least one check. Tokyo matches both checks, so it shows up only once.

```sql
-- Show cities that are not Delhi or Sao Paulo, or cities named Tokyo
SELECT
    name, area
FROM
    cities
WHERE
    name NOT IN ('Delhi', 'Sao Paulo') OR
    name = 'Tokyo';
```

Figure 10 — NOT IN ... OR ...

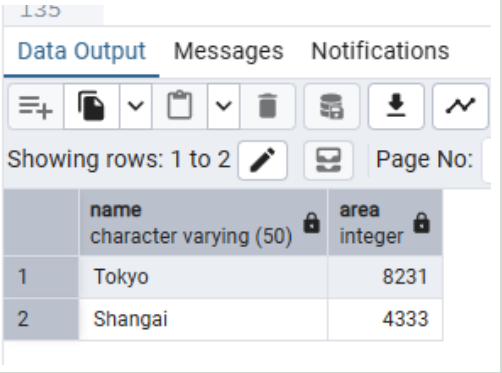


Figure 11 — Result: Tokyo and Shangai

7. Summary

Operator	Remember
BETWEEN a AND b	Is the value between the lower bound a and the upper bound b?
IN (...)	Is the value in the list? The list can have strings or numbers
NOT IN (...)	Is the value not in the list? The opposite of IN
AND	The row must match both checks
OR	The row must match at least one check
Chaining	We can join as many checks as we want with AND and OR

Next lesson: a quick exercise.

Sources
Lecture transcript and lecture image (Figure 1)
My own practice screenshots (Figures 2–11)